

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

November-2013

MA 203: Engineering Mathematics-IV

Date: 11.11.13, Monday

Time: 01:30 p.m. To 04:30 p.m.

Total: 70 Marks

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(4+3)

- (a) Find Fourier Sine Transform of the function $f(x) = \frac{e^{-ax}}{x}$
- (b) Find Fourier Cosine Transform of the function $f(x) = e^{-2x} + 4e^{-3x}$

Q-2

(4+5+5)

- (a) Define: Analytic function and orthogonal system. Show that every analytic function $f(z) = u + iv$ defines two families of curves $u(x, y) = 0$ and $v(x, y) = 0$ which forms an orthogonal system.
- (b) In two dimensional fluid flow, the velocity potential is given by $\phi = x^2 - y^2$ find flux function ψ . Is ϕ a Harmonic function?
- (c) Under the transformation $w = z^2$, find the image in w-plane of square at points $(0,0)$, $(2,0)$, $(2,2)$, $(0,2)$ also sketch the image.

OR

Q-2

(4+5+5)

- (a) Show that an analytic function with constant modulus is constant.
- (b) State Cauchy-Riemann equations in polar form also find value of k such that the function $f(z) = r^2 \cos 2\theta + ir^2 \sin k\theta$ be an analytic function.
- (c) Find the image of $|z - 3i| = 3$ under the mapping $w = \frac{1}{z}$

Q-3

(3+4+3+4)

(a) By method of Residues, obtain the inverse Laplace transform of $\frac{s+1}{s^2+2s}$

(b) Evaluate: $\int_C \frac{z^3 - z}{(z-2)^3} dz$, where C is the circle $|z|=3$

(c) Evaluate: $\int_0^{1+i} (x^2 + iy) dz$ along the path $y = x^2$

(d) Define Harmonic function; show that $u(x, y) = \frac{1}{2} \log(x^2 + y^2)$ is Harmonic function.

OR

Q-3

(5+5+4)

(a) Find the Taylor series expansion for $f(z) = \cos z$ at point $z = \frac{\pi}{4}$

(b) State Cauchy residue theorem and hence evaluate: $\int_{C: |z|=\frac{3}{2}} \frac{4-3z}{z(z-1)(z-2)} dz$

(c) Obtain Laurent's series expansion of function $f(z) = \frac{1}{z^2 + 4z + 3}$ in the regions $|z| < 1$ and $|z| > 3$

SECTION-II

Q-4

(3+4)

(a) Define unit step function and find its Z-transform

(b) Define Dirac delta (unit impulse) function and find its Z-transform

Q-5

(5+5+4)

(a) Evaluate $\int_0^1 e^{-x^2} dx$ by the Gauss integration formula for $n=3$

(b) Find the interpolating polynomial for (0, 2), (1, 3), (2, 12) and (5, 147) using Langrange's interpolation formula.

- (c) Compute $\cosh(0.56)$ using Newton Forward difference formula for the following table

x	0.5	0.6	0.7	0.8
$f(x)$	1.127626	1.185465	1.255169	1.337435

OR

Q-5

(7+7)

- (a) Evaluate $\int_0^1 e^{-x^2} dx$ using (i) Trapezoidal rule and (ii) Simpson's $\frac{1}{3}$ rd rule taking $h = 0.2$

- (b) Use Newton-Raphson method to solve the transcendental equation: $3x = \cos x + 1$

Q-6

(7+7)

- (a) Find by false position method the real root of the equation $\log_e x = \cos x$

- (b) Apply the fourth order Runge-Kutta method to solve:

$$\frac{dy}{dx} = xy, \text{ for } x = 1.2 \text{ initially, } y(1) = 2. \text{ Take step size } h = 0.2$$

OR

Q-6

(7+7)

- (a) Using Picards method, solve the differential equation: $\frac{dy}{dx} = 1 + xy$, $y(0) = 1$ for $x = 0.1$

- (b) Find a real root of the equation $x^3 + x^2 - 1 = 0$ using bisection method.
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